SolidJS: Is SolidJS actually faster and better than React?

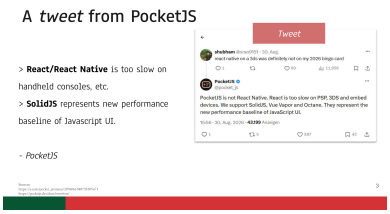
Adolfo Loewenthal

07-11-26

Origin of interest

A tweet from PocketJS

- PocketJS is a framework to create JavaScript UI apps on handheld consoles [1]
- A post from their social media account caught my attention
- First important detail: React/React Native is too slow for handheld consoles
- Second important detail: SolidJS represents new performance baseline of Javascript UI

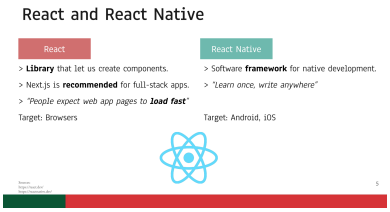


[1] PocketJS. Documentation. `https://pocketjs.dev/docs/overview/`

What I was using is slow?

React and React Native

- At that time I was using Next.js for a project, which is React based
- Therefore, I thought that what I was using could be slow
- Regarding React, we can define it as a library, which let us create components. The target of React is browsers. [2]
- On the other hand, React Native is a framework for native development. The target of React Native is Android and iOS. [3]



[2] React. Documentation. `https://react.dev/`

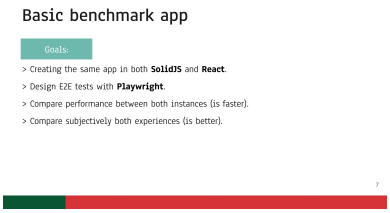
[3] React Native. Documentation. `https://reactnative.dev/`

Was my curiosity still valid?

PocketJS was talking about handheld consoles, which isn't my target platform on real life. That could mean, my comparison might be totally out of place. However, I set some goals to measure the performance difference between SolidJS and React.

Basic benchmark app

- The proposed app should be coded in both SolidJS and React, using the same patterns
- With help of Playwright, I designed some E2E tests to automatize the performance measure



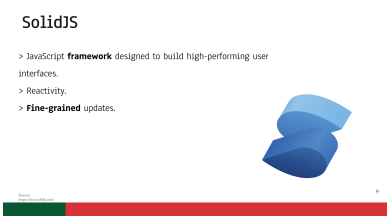
- If we compare the performances after running the tests, I can subjectively say which one is faster (if my test is reliable)
- Because I wrote both apps from scratch, I would determine which setup process and learning curve I enjoyed more. This reflection could be used to say which one is better.

What is SolidJS?

SolidJS [4]

- It is a JavaScript framework designed to build high-performing UIs
- Just like React, it also features reactivity
- The main feature of SolidJS are their fine-grained updates

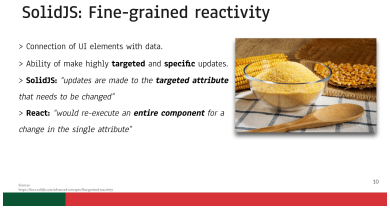
[4] SolidJS. Documentation. `https://docs.solidjs.com/`



SolidJS: Fine-grained reactivity [5]

- Fine-grained reactivity is a method of handling updates of data in components
- In short, it is the ability of make highly targeted and specific updates
- SolidJS would only update the targeted attribute and parts connected to it
- React re-renders an entire component just for the same need

[5] SolidJS. Fine-grained reactivity. `https://docs.solidjs.com/advanced-concepts/fine-grained-reactivity`

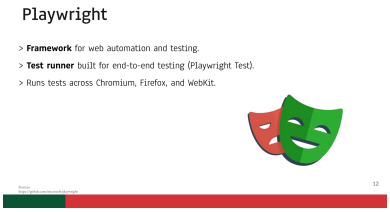


What is Playwright?

Playwright [6]

- It is a framework used for web automation and testing
- Specifically, the test runner Playwright Test can perform end-to-end testing
- With help of Playwright, we can measure the performance from a web application

[6] Playwright. Documentation. `https://playwright.dev/`



Setting up experience

SolidJS 2.0 (RC) + Vite

- It has an interactive installer, which helps to choose the configuration



- The docs are a bit raw because SolidJS 2.0 (RC) is relatively new



React Router + Vite

- It also has an interactive installer
- The docs are very good and complete

Code comparison

SolidJS and React

The two featured images in the slides show the same component using both SolidJS and React respectively.



- The code is practically the same
- The main difference is the use of SolidJS and React helpers
- In SolidJS, we use **createSignal()** and **createStore()** for single values and objects or arrays respectively
- In React, we can cover both instances with **useState()**

Pros and cons

Pros (SolidJS)

- Code is easy to migrate from and to React
- Easy interactive installation
- Fine-grained reactivity

Cons (SolidJS

- In transition to Solid 2.0 RC (13.08.26), so docs are still raw [7]
- It's still a niche framework

Pros (React)

- Code is easy to migrate from and to SolidJS
- Easy interactive installation
- Top 2 most popular web frameworks globally (+40%) [8]

[7] SolidJS.. Solid 2.0 rc: The big <reveal>. `https://www.solidjs.com/blog/solid-2-0-rc-the-big-reveal`

[8] Statista. Most used web frameworks among developers worldwide as of 2026. `https://www.statista.com/statistics/1124699`

Cons (React

- Re-renders components by default when updating
- Virtual DOM + Real DOM to compare changes [9]

[9] Nathan Babcock. Solidjs vs. react: The go-to guide. https://www.toptal.com/developers/react/solidjs-vs-react

References

Nathan Babcock. Solidjs vs. react: The go-to guide. https://www.toptal.com/developers/react/solidjs-vs-react.

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SolidJS. Documentation. https://docs.solidjs.com/.

SolidJS.. Solid 2.0 rc: The big <reveal>. https://www.solidjs.com/blog/solid-2-0-rc-the-big-reveal.

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